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Enhancement of photovoltaic module performance using passive cooling (Fins): A comprehensive review

Hussein A. Kazem [a,b], AliH.A. Al-Waeli [c], Miqdam T. Chaichan [d], K. Sopian [e], Al-Amiery Ahmed [d,f], Wan Isahak Wan Nor Roslam [d,f,*]

[a] Sohar University, PO Box 44, Sohar, PCI 311, Oman
[b] Solar Energy Research Institute, Universiti Kebangsaan Malaysia, 43600, Bangi, Selangor, Malaysia
[c] Engineering Department, American University of Iraq, Sulaimani, Kurdistan Region, Sulaimani, Iraq
[d] Energy and Renewable Energies Technology Research Center, University of Technology, Iraq
[e] Department of Mechanical Engineering, Universiti Teknologi PETRONAS, 32610 Seri Iskandar, Perak Darul Ridzaun, Malaysia
[f] Department of Chemical and Process Engineering, Faculty of Engineering and Built Environment, Universiti Kebangsaan Malaysia (UKM), Bangi, 43600, Malaysia

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ABSTRACT

Photovoltaic-thermal (PV/T) technology, combines the benefits of both solar photovoltaic (PV) and solar thermal systems into a single integrated solution. It is a promising renewable energy technology that maximizes solar energy utilization and offers multiple benefits for sustainable power generation. A review and discussion of both active and passive thermal management solutions for PV technology is included in this paper. Using fins on the back of PV is the focus of this paper in terms of its design and performance parameters. Fins are used in air, fluid, nanofluid and PCM cooling PV/T systems. Installing them in these systems improves their performance and increases their electrical and thermal efficiency together in rate range between 8 and 10%. Furthermore, this article examines different approaches to designing and implementing systems, as well as how to optimize results, in addition to exploring potential problems and opportunities. Aiming at creating, manufacturing, utilizing, and potentially promoting high-performance photovoltaic-based technologies and systems which include fins are the purpose of this article.

1. Introduction

After World War II, there was a growing recognition of the need to expand the use of conventional energy sources based on fossil fuels. The devastation caused by the war, along with growing environmental concerns, economic considerations, and energy security issues, has led to a global shift toward renewable energy use sources. One of the main drivers of the transition was the realization of the finite nature of fossil fuels and the need for long-term energy sustainability [1]. The heavy reliance on coal, oil and natural gas during the post-war period highlighted its limitations and vulnerabilities. After the seventies of the last century, it became clear that a diversified and flexible energy system is necessary to ensure stability and independence.

Environmental concerns have also played an important role in promoting the adoption of renewable energies. The world has witnessed large-scale industrial activities and unprecedented levels of pollution, which has led to environmental degradation. Recognition of the harmful effects of burning fossil fuels, such as air pollution and greenhouse gas emissions that contribute to climate

* Corresponding author. Energy and Renewable Energies Technology Research Center, University of Technology, Iraq.
E-mail address: wannorroslam@ukm.edu.my (W.I. Wan Nor Roslam).

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change, has increased the urgency of finding cleaner and more sustainable alternatives. In addition, the economic benefits of renewable energies are becoming apparent [2,3].

Investment in renewable energy technologies has provided opportunities for job creation, technological advancement, and economic growth. Governments and private institutions have recognized the potential for innovation and competitiveness in the emerging renewable energy sector. Moreover, the geopolitical landscape has shaped the desire to reduce dependence on imported fossil fuels. Countries have sought energy independence and security by diversifying energy sources, reducing exposure to political conflicts, price fluctuations, and supply disruptions associated with fossil fuels. The development of domestic renewable energy sources provided a means to achieve energy self-sufficiency [4,5].

Since the 1973 war and the resulting reduction in oil imports, the urgent need to switch to renewable energies has increased. The accelerating effects of climate change, growing energy demand, and declining costs of renewable technologies have further reinforced the need for a clean and sustainable energy transition. Efforts are being made globally to promote renewable energy deployment, improve energy efficiency, and strengthen international cooperation to mitigate the environmental impacts of energy production and consumption. In short, the need to shift to renewable energies after World War II stemmed from concerns about long-term energy sustainability, environmental degradation, economic opportunity, and energy security. Transitioning to renewable energies has become crucial to addressing these challenges and building a more sustainable and resilient future [6,7].

Solar energy is a renewable energy source that harnesses the energy of sunlight to generate electricity, produce distilled water, and provide thermal energy. It provides a sustainable and reliable alternative to traditional fossil fuel-based energy systems, with a wide range of applications.

Solar energy can be used to produce distilled water through a process called solar desalination. This application is especially valuable in areas where access to clean water is limited. Solar desalination systems use solar thermal energy to evaporate water and then collect the vapour that condenses, leaving behind salts and impurities. This process provides a sustainable solution to water filtration and can help address water scarcity issues [8,9].

Another important application of solar energy is in thermal heating systems. Solar thermal collectors capture the sun's thermal energy and use it to heat water, air, or other liquids. Solar thermal systems are commonly used for domestic water heating, space heating in buildings, and even industrial processes that require heat. By utilizing solar thermal energy, these applications reduce reliance on fossil fuel-based heating systems and reduce greenhouse gas emissions [10,11].

The primary application of solar energy is in the generation of electricity through photovoltaic (PV) systems. Solar panels with photovoltaic cells convert sunlight directly into electricity using the photovoltaic effect. This clean, sustainable production of electricity can be used to power homes, businesses, and even entire communities [12]. Solar PV systems can be installed on rooftops, on solar farms, or integrated into various structures to harness the abundant solar energy available. In addition to electricity generation, water distillation and thermal heating, solar energy finds applications in various other sectors. Solar energy can power off-grid systems such as solar street lights, telecom infrastructure, and off-grid homes or cabins [13]. It can also be used for solar powered pumps in irrigation systems, providing a sustainable water supply for agriculture [14]. Moreover, solar energy plays an important role in operating transportation. Solar-powered vehicles, including solar-powered cars, buses, and boats, use photovoltaic panels to convert sunlight into electricity for propulsion. Solar energy can also be used to charge electric vehicles, reducing carbon emissions from transportation [15,16].

Overall, solar energy provides a versatile and sustainable solution for stable electricity production, water distillation and thermal energy. Their applications span a wide range of sectors, from residential and commercial electricity generation to water purification, heating systems, off-grid power supplies, and transportation. As solar energy technologies continue to advance and become more cost effective, solar energy is expected to play an increasingly important role in our transition to a clean and sustainable energy future.

2. High temperatures can have several effects on the current and voltage output of photovoltaic (PV) modules

Current Reduction: High temperatures can cause a reduction in the current output of PV modules. This is primarily due to an increase in the internal resistance of the solar cells. As the temperature rises, the resistance within the cells increases, impeding the flow of current. The increased resistance leads to a decrease in the current generated by the PV module [17].

Voltage Reduction: The voltage output of PV modules is also affected by high temperatures. Similar to the current reduction, the increase in internal resistance causes a drop in voltage. As the temperature rises, the voltage at the output terminals of the PV module decreases. This reduction in voltage can impact the overall performance and power output of the PV system [18].

Power Loss: The combination of reduced current and voltage output due to high temperatures leads to a decrease in the overall power output of the PV modules. The power output of a PV module is the product of its current and voltage. As both current and voltage decrease under high-temperature conditions, the power output is significantly reduced [19]. This power loss affects the efficiency and performance of the PV system [20].

Efficiency Decrease: The efficiency of a PV module is a measure of how effectively it converts sunlight into electrical energy. High temperatures can cause a decrease in the efficiency of PV modules. The reduction in current and voltage output, as well as the associated power loss, contribute to a decline in the overall efficiency of the PV system. The decrease in efficiency affects the energy yield and performance of the system [21]. As a general rule, photovoltaic cells perform best at cooler temperatures and degrade as they warm up [22]. Using cooling fluids such as air or liquids, the researchers were able to design and build several systems that cooled photovoltaic modules. The accumulated heat is dissipated by forced air movement (using air intake fans) on the surface of PV panels that use air as a cooling fluid. Cooling fluids such as water or nanofluids absorb the heat accumulated in the system and transfer it away through a circulation system. Liquid cooling generally outperforms air cooling when comparing the two methods. In comparison to the

photovoltaic module, the PV cooling system (PV/T) leads to higher electrical productivity and thermal efficiency [23,24]. Also, this cooling of the PV module will extend the life of the unit for an additional period. There are also systems that work with passive cooling, which is the cooling of the PV panels using convection and radiation without the help of any additional devices. These panels are designed in such a way that they can absorb the accumulated heat and then dissipate it to the surroundings by themselves, without any external assistance. This dissipation process helps lower the temperature of the PV panels, which helps them increase their efficiency and operational life [25,26].

A review of the latest studies on finned PV modules is presented in this paper. However, a brief overview of different cooling methods used in PV/T system is provided. An analysis and comparison of fin design and configuration was conducted. Additionally, fin cooling was compared to other cooling methods. The study aims to present the juices of what researchers have accomplished in this field. The purpose is to present it to researchers and developers in order to improve what has been done and to produce the best devices at the lowest cost in the future.

3. Effect of temperature on photovoltaic module

PV modules are highly sensitive to temperature. The power output of PV modules decreases as their temperature increases. A decrease in light energy conversion to electricity is responsible for this [27]. Moreover, a higher temperature increases the electrical resistance of the cells, causing the current generated to decrease. Additionally, the module's power output is typically negatively correlated with temperature, which means that as the temperature increases, the power output decreases [28]. As a result of thermal stress on the cells, high-temperature operation of a PV module can also decrease the module's life expectancy. PV module efficiency decreased by 5% for every 10 °C increase [29]. PV module temperatures rise due to internal losses and external environmental conditions. Hybrid PVT collectors and cooling systems have been designed to counteract this effect. PVTs are thermal collectors mounted on the back of PV modules, which enhance heat transfer and maximize thermal power. As the PV module's temperature rises, heat transfer increases, providing cooling as well [30].

Solar cells and photovoltaics are mathematically represented by the one-diode model, an approach that assumes each cell contains a single diode connected in parallel with an electrical source [31]. In the model, five parameters are involved: the diode ideality factor (n), saturation current (Io), series resistance (Rs), shunt resistance (Rsh), and photocurrent (Iph). In order to determine the parameters, the model is fitted to the measured current-voltage (I–V) characteristics of the solar cell. Under different light intensities and temperatures, the following equation can be used to predict a solar cell's behaviour:

$$I_L = I_{ph} - I_o \left\{ e^{\left[\frac{q\,(V+I_L R_s)}{n \times k \times T} \right]} - 1 \right\} - \frac{V + I_L R_s}{R_{sh}} \tag{1}$$

The electric power of the PV is expressed as:

$$P_{max} = I_{mp} \times V_{mp} \tag{2}$$

Where (I_{mp}) is the current and (V_{mp}) is the voltage. The electrical efficiency (η_e) of conventional PV is calculated using the formula below:

$$\eta_e = \frac{P_{max}}{G \times A_{module}} \tag{3}$$

Where "(G) is the solar irradiance in W/m^2 and (A_{module}) is the PV module area in m^2".

PV module temperature is influenced by two coefficients:

1. Coefficient of PV Temperature: The following equation estimates the decrease in efficiency of a PV module as temperature increases. PV module electrical efficiency is temperature-dependent, η_e the expression is given as below:

$$\eta_e = \eta_r \left(1 - \beta * \left(T_{pm} - T_r\right)\right) \tag{4}$$

1. COT (Coefficient of Temperature) of Power Outputs: As the temperature increases, this equation calculates the power output of a PV module. It is expressed as:

$$P_{Tc} = \eta_r (1 - \beta * (T_c - T_r)) \times G \times A \tag{5}$$

Where "T_c is the PV operating temperature. As the temperature of a cell increases, its open circuit voltage (V_{oc}) decreases and its short circuit current (I_{sc}) increases". Moreover, open circuit voltage is usually negatively correlated with temperature, while short circuit current is positively correlated with temperature. As the temperature rises above 25 °C, the open circuit voltage decreases and the short circuit current increases. Cells operating at their highest operating temperature (HOT) experience the greatest decrease in open circuit voltage and the greatest increase in short circuit current. By using the appropriate electrical configuration, the increase in short circuit current can be exploited by minimizing the decrease in open circuit voltage due to temperature. Using a higher short circuit current can increase the power output of the module, for example, if the temperature is expected to be high [32].

4. Photovoltaic cooling methods

PV modules can be cooled passively (natural convection) or actively (forced convection) as shown in Fig. 1(a).

1. Natural Convection: This is the simplest and most common method of cooling PV modules. During natural convection, warm air rises as a result of the PV module's heat and is replaced by cooler air [33].
2. Forced Convection: By forcing air over the PV module, the convection process is increased and cooling efficiency is improved [34].
3. Heat Sink: The heat sink is a metal plate that absorbs and disperses heat. It is possible to use heat sinks in conjunction with natural or forced convection [35].
4. Phase Change Materials: Thermal storage materials absorb heat from PV modules and prevent them from overheating [36].
5. Liquid Cooling: In this method, heat is absorbed and dissipated from a PV module with a liquid instead of air. It is used in applications that require a very high level of power [37].

Fig. 1(b) illustrates literature-tested and investigated PV module cooling methods [38,39].

Photovoltaic panels can be cooled naturally, by forced convection, actively, or passively. It is most economical to cool naturally by allowing the air to flow naturally. Heat is transferred via forced convection when fans are used to draw air across the panel. A liquid coolant, such as water or glycol, is used to cool an active cooling panel. The heat from the panel is dissipated passively by radiative and conductive heat transfer. The cost, efficiency, and simplicity of a cooling approach should be considered when choosing an appropriate approach for analysis [40]. While natural cooling is the most economical option, high-power photovoltaic panels may not be able to handle it. In comparison with natural cooling, forced convection is more efficient but requires more equipment and energy to operate [41,42]. A large investment is required for active cooling, which is the most efficient. As the simplest cooling method, passive cooling requires little equipment, but it is less efficient than other cooling methods. Considering these factors, the best way to assess the thermal performance of a heat sink will depend on the application's specific requirements. Several cooling approaches should be evaluated to determine which is most effective.

4.1. Photovoltaic thermal water cooling

The photovoltaic thermal water cooling system utilizes both photovoltaic (PV) cells and thermal energy to heat water. Using PV cells, the sun's energy is converted into electricity, while thermal energy is used to heat the water. As shown in Fig. 2(a), the photovoltaic cells produce energy that drives a pump to move water through a heat exchanger [43]. Water becomes warmer after passing through the heat exchanger, which absorbs the heat from the sun. Heat can also be used to heat pools and for industrial purposes in addition to space heating and domestic water heating.

4.2. Photovoltaic thermal air cooling

This type of HVAC system uses solar energy to cool buildings. PV/T air cooling (PVTAC) has become increasingly popular in recent years. According to Fig. 2(b), PVTAC systems generate electricity through the use of photovoltaic (PV) panels, which then propel a fan to circulate air through an adsorption chiller [44]. By cooling the air, conventional air conditioning systems are reduced in necessity. It can help reduce energy consumption and carbon emissions by replacing traditional cooling systems with PVTAC systems.

4.3. Photovoltaic spray cooling

A spray cooling system for photovoltaics reduces the operating temperature of solar cells and modules while improving their

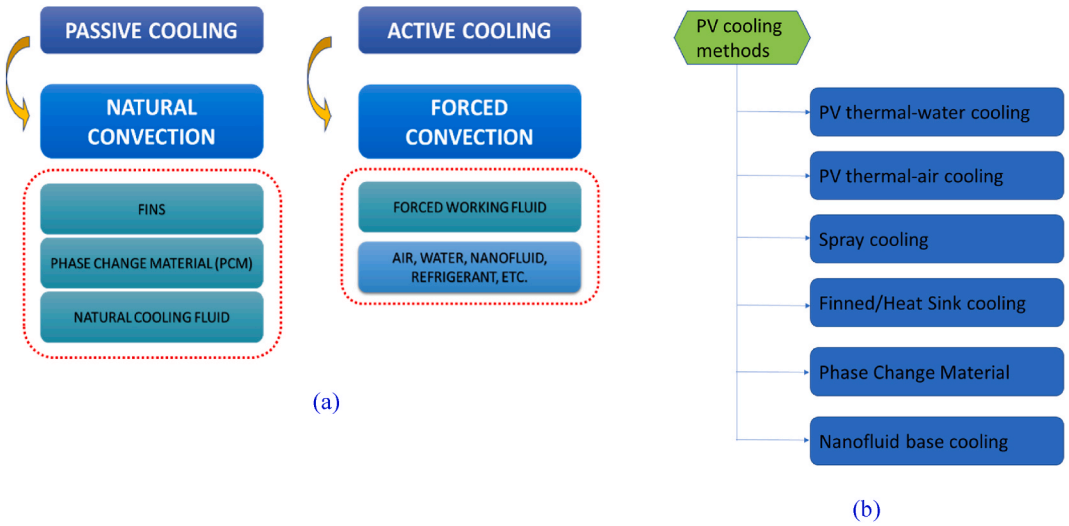


Fig. 1. PV cooling based on (a) convection, (b) methods.

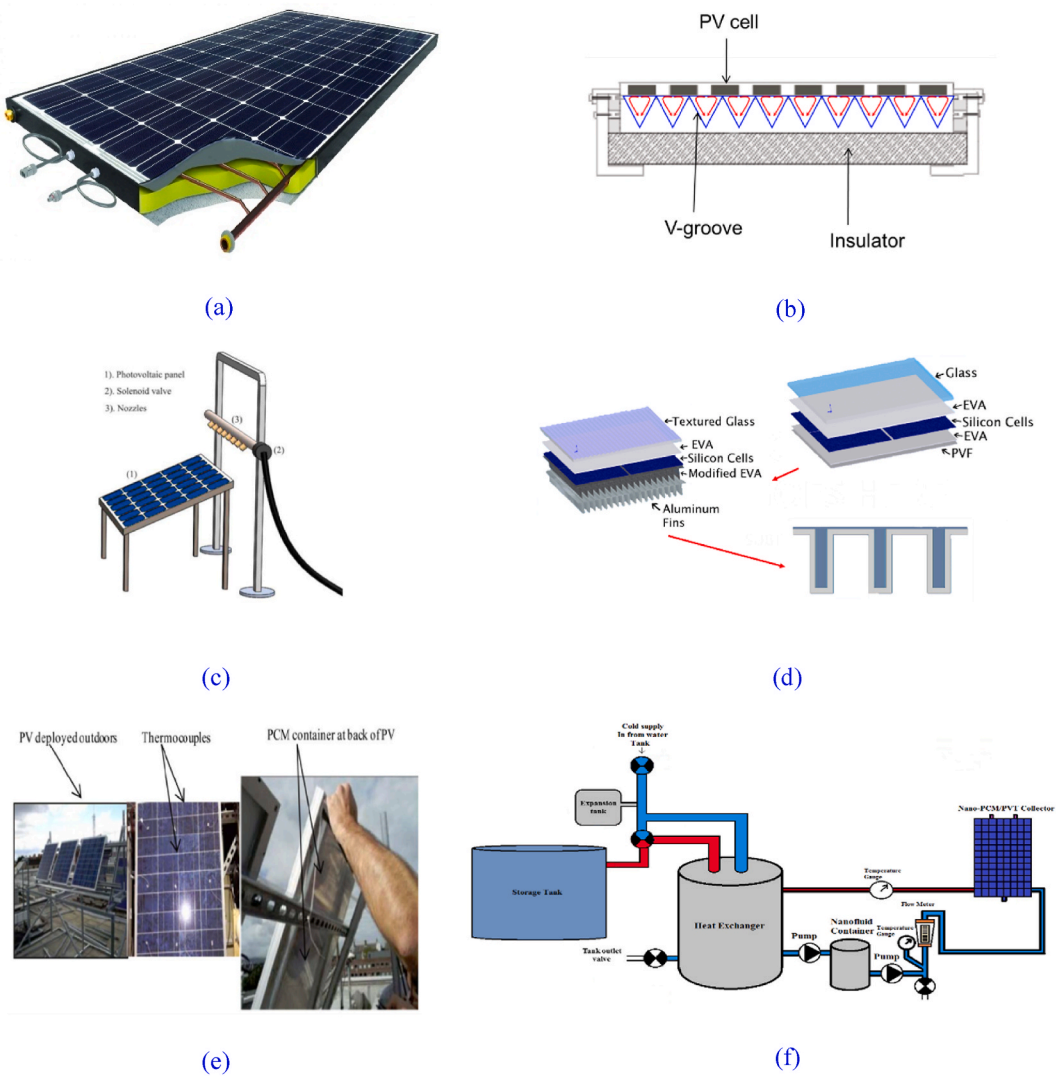


Fig. 2. Cooling methods (a) water, (b) air, (c) spray, (d) fins, (e) PCM and (f) nanofluid [56,57].

efficiency. Fig. 2(c) illustrates how water is sprayed over solar panels in order to absorb heat generated by the cells [45]. In addition to air cooling or fan cooling, this method can also be used as a stand-alone cooling solution. In order to generate the mist, either spray nozzles or pressurized systems must be used.

4.4. Fin/heat sink cooling

According to Fig. 2(d), fins on a heat sink are used to dissipate heat from photovoltaic arrays [22]. Fins allow the heat sink to absorb and dissipate more heat by increasing its surface area. Photovoltaic arrays can use this cooling technique in hot climates, since the additional surface area keeps them cool and efficient.

4.5. Phase change material cooling

Phase change materials (PCMs) are substances that can undergo a phase transition, typically from solid to liquid or vice versa, while absorbing or releasing a significant amount of latent heat. These materials have the unique property of storing and releasing thermal energy during phase transitions, making them valuable in various applications [46].

Phase change materials include organic and inorganic compounds. Organic PCMs are typically hydrocarbons or fatty acids, such as paraffin wax or polyethylene glycol. Inorganic PCMs include salts like sodium sulfate or hydrated salts like calcium chloride hexahydrate. Each type of PCM has different melting and solidification points, latent heat capacities, and other characteristics that make them suitable for specific applications. Phase change materials are used in cooling photovoltaic (PV) modules. PV modules generate electricity from the sunlight but experience efficiency losses due to high operating temperatures. Excessive heat can reduce the modules' output power and lifespan. PCMs can mitigate these issues and improve PV system performance [47].

When integrated with PV modules, PCMs act as a thermal energy storage medium. During periods of high solar radiation or ambient temperature, the PCM absorbs excess heat from the module. This causes it to undergo a phase transition from solid to liquid. This process effectively stores excess thermal energy within the PCM. As the ambient temperature decreases or solar radiation diminishes, the PCM solidifies again, releasing stored heat back into the surroundings. This heat transfer process regulates PV module temperature and prevents overheating [48].

By incorporating PCMs into PV systems, thermal management is enhanced, leading to increased energy production and improved reliability. PCMs can also reduce dependency on active cooling systems, such as fans or liquid cooling, reducing energy consumption and maintenance requirements. Overall, phase change materials offer a promising solution for cooling PV modules by efficiently managing thermal energy generated during operation. Their ability to store and release heat during phase transitions makes them an effective passive cooling strategy, enhancing photovoltaic systems' performance and longevity [49].

An examination of the ways in which PV/T systems can be cooled using water as a cooling fluid for a heat storage phase change material was conducted by Preet [50]. Photovoltaic panels can be effectively cooled using this method, according to the review. According to the author, phase change materials have low thermal conductivity and harden at night, affecting their readiness when used. To improve PCMs' thermal conductivity, nanomaterials were suggested. Additionally, Preet considered that the constant extraction of heat from the PCM would maintain the storage capacity of the phase change material. The PCM in PV/T systems absorbs heat generated by the panel and releases it when the temperature drops, as shown in Fig. 2(e) [51]. By reducing the panel's temperature, the system's efficiency is improved.

4.6. Nanofluid base cooling

Nanofluids are a class of engineered fluids that consist of a base fluid, such as water or oil, infused with nanoscale particles called nanoparticles. These nanoparticles, typically with sizes ranging from 1 to 100 nm, are dispersed within the base fluid, creating a stable suspension. The addition of nanoparticles to the fluid enhances its thermal and heat transfer properties, making nanofluids promising for various applications, including cooling photovoltaic (PV) modules [52].

There are different types of nanoparticles used in nanofluids, such as metallic nanoparticles (e.g., copper, silver), metal oxide nanoparticles (e.g., aluminium oxide, titanium dioxide), and carbon-based nanoparticles (e.g., carbon nanotubes, graphene). Each type of nanoparticle has unique properties that influence the heat transfer characteristics of the nanofluid [53]. When applied to cooling PV modules, nanofluids offer improved heat dissipation capabilities compared to traditional cooling fluids. The presence of nanoparticles in the base fluid significantly enhances thermal conductivity, allowing for more efficient transfer of heat away from the PV module. This improved thermal performance helps to mitigate the temperature rise of the PV cells, which can lead to reduced efficiency and accelerated degradation [54]. Nanofluids can be circulated through the PV module using a closed-loop cooling system or by direct immersion. In a closed-loop system, the nanofluid absorbs heat from the PV module and transfers it to a heat exchanger, where it dissipates into the surroundings. This continuous circulation of the nanofluid helps maintain the PV module at an optimal operating temperature [55]. The use of nanofluids in PV module cooling offers several advantages. The enhanced thermal conductivity of nanofluids improves heat transfer efficiency, resulting in lower PV cell temperatures and improved overall performance. Additionally, nanofluids can potentially reduce the size and complexity of cooling systems, as they provide superior cooling capabilities compared to conventional fluids [56]. However, it is important to consider certain factors when using nanofluids, such as nanoparticle stability, sedimentation, and the potential for particle agglomeration over time. These aspects can impact the long-term performance and reliability of the cooling system [57].

In summary, nanofluids are advanced heat transfer fluids that incorporate nanoparticles into a base fluid, offering enhanced thermal conductivity. By utilizing nanofluids for cooling PV modules, the heat dissipation capabilities can be significantly improved, leading to lower operating temperatures, increased energy production, and prolonged lifespan of the modules. Fig. 2(f) shows a system for cooling photovoltaic cells with nanofluids as the cooling medium.

4.6.1. Passive cooling by fins

Fins are an essential component of various devices and systems designed to improve fluid or air flow. They are commonly used in applications such as cooling systems, propulsion systems, and aerodynamics [58]. Here's an overview of the benefits of fins, the materials they are made of, and the optimal conditions for their operation.

4.6.2. Benefits of fins

Heat Dissipation: Fins are often used to enhance heat dissipation by increasing the surface area available for heat transfer. They can be found in various cooling systems, such as radiators in cars, heat sinks in electronics, and air conditioning units [59].

Improved Propulsion: In the realm of fluid dynamics, fins can be used to improve propulsion. For example, fins are employed in marine vehicles like submarines, ships, and torpedoes to enhance manoeuvrability and stability by providing directional control [60].

Aerodynamic Efficiency: Fins play a crucial role in aerodynamic designs, particularly in aviation and automotive industries. By manipulating airflow around an object, fins can reduce drag, improve stability, and enhance overall efficiency [61].

4.6.3. Materials used for fins

The choice of materials for fins depends on the specific application and desired properties. Common materials used for fins include:

Metals: Aluminium and copper are frequently employed due to their excellent thermal conductivity. These metals efficiently transfer heat and are commonly found in heat sinks and radiators [62].

Composites: Composite materials, such as carbon fibre reinforced polymers (CFRP), offer a balance between strength, weight, and thermal properties. They are commonly used in high-performance applications, including aerospace and sports equipment [63].

Plastics: Certain thermoplastics, like polyethylene and polypropylene, are suitable for fins in applications where weight and cost are critical factors. Plastic fins are commonly used in air conditioning units and other consumer products [64].

4.6.4. Optimal conditions for Fin operation

To ensure the optimal performance of fins, several factors should be considered:

Surface Area: Fins with larger surface areas offer increased heat transfer or aerodynamic capabilities. However, there is a trade-off between surface area and other factors like weight, cost, and space constraints [65].

Geometry: The design of the fin, including its shape, thickness, and spacing, can significantly impact its performance. The geometry should be optimized to minimize resistance or maximize heat transfer, depending on the application [66].

Fluid or Airflow: The efficiency of fins depends on the flow rate and characteristics of the fluid or airflow. Higher flow rates generally enhance heat transfer or aerodynamic performance. However, factors like pressure drop, turbulence, and boundary layer effects should be considered during the design process [67].

Maintenance and Cleaning: Fins can accumulate dust, debris, or contaminants over time, which can reduce their efficiency. Regular maintenance and cleaning are essential to ensure optimal operation [68].

It's important to note that the specific requirements and optimal conditions for fins can vary depending on the application, industry, and engineering considerations. Consulting domain experts or relevant literature can provide further insights into the specific requirements for a particular use case.

PV/T systems combine the advantages of both solar thermal collectors and photovoltaic modules. Fins play an important role in PV/T systems by improving their thermal performance [69]. Here is an overview of the optimal use of fins in PV/T systems:

Improve heat transfer: Fins are used in PV/T systems primarily to improve heat transfer from the photovoltaic cells to the circulating fluid. By increasing the surface area available for heat transfer, the fins enhance the overall thermal efficiency of the system [70].

Fin Design: Fin design in a PV/T system is critical to optimal performance. Factors to consider include fin material, shape, size, thickness, and arrangement. Fin geometry must be optimized to maximize heat transfer while minimizing pressure drop and parasitic heat loss [71].

Material selection: The fins in PV/T systems are usually made of materials with high thermal conductivity to facilitate efficient heat transfer. Common materials include metals such as aluminium or copper, which provide excellent thermal properties. Material selection should take into account factors such as cost, durability, and compatibility with the working fluid [72].

Fin Orientation: The orientation and position of the fins in a PV/T system is important for efficient heat transfer. The fins are usually arranged perpendicular to the incident solar radiation to increase heat absorption. The spacing between the fins must be optimized to balance heat transfer and airflow resistance [73].

Fluid flow: Proper fluid flow is essential for efficient heat transfer in PV/T systems. The flow rate, velocity, and distribution across the fins must be optimized to ensure efficient cooling of the photovoltaic cells. Adequate fluid circulation helps maintain low cell temperatures, and improves electrical and thermal performance [74].

Cleaning and Maintenance: Regular cleaning and maintenance of the fins are essential to ensure optimal performance in PV/T systems. Accumulation of dust, dirt or debris on the fin surfaces can impede heat transfer and reduce system efficiency. Cleaning protocols should be put in place to keep the finches clean and free of obstructions [75].

Integration with photovoltaics: Fins must be properly integrated with photovoltaics to increase thermal contact and heat transfer. The direct contact between the fins and cells allows for efficient heat dissipation. Appropriate thermal interface materials or technologies can be used to enhance thermal coupling and reduce thermal resistance [76].

System Monitoring and Control: Monitoring the temperature and performance of photovoltaics, fluid flow rate, and overall system efficiency is essential for optimal fin utilization in PV/T systems. Control strategies can be implemented to optimize system operation, such as modulating flow rates, optimizing fan speed, or implementing algorithms Intelligent control [77].

It is important to note that the optimal use of fins in PV/T systems can vary depending on the specific design, climatic conditions, and intended application. Consulting experts in the field or referring to the specialized literature and research papers can provide more detailed guidance and specific applications [25,26].

According to Venkateswarlu et al. [77], several studies have enhanced the use of monomeric nanofluids (the addition of one nanomaterial to a base fluid). In addition, the authors reviewed the most advanced nanofluids, such as hybrid and triple nanofluid cooling PV/T systems. In conclusion, the field remains open to exploring new nanomaterials with the potential to improve PV/T systems through advanced techniques.

In their study, Micheli et al. [78] investigated the feasibility of using micro-fin plates in natural convection to cool concentrated photovoltaic (CPV) devices. It was demonstrated by the authors that plate micro-fins can significantly enhance cooling efficiency for CPV systems through various experiments. CPV systems can be passively cooled effectively and efficiently with this approach.

PV panels were designed in a novel manner by El Mays et al. [79]. A finned aluminium plate is proposed to cover the panel to increase surface area, thereby enhancing cooling. In order to study how the finned plate affected the performance of the panel, the authors conducted experiments. In addition to improving efficiency, the finned plate reduced cooling costs by up to 11%. A finned plate also had little impact on the panel's appearance, according to the authors. This design has the potential to improve the productivity of PV cells, thereby reducing the cost of generating energy.

Using multi-level fin heat sinks (MLFHS), Ahmad et al. [80] assessed photovoltaic cooling efficiency. MLFHS thermal dissipation was analyzed using numerical simulations by reducing the fin elevation under natural convection. Heat dissipation from the MLFHS is increased when the fin height is truncated. The authors also demonstrated that the fin height and number of fins can be adjusted to increase the rate of heat dissipation in the MLFHS. It appears that the MLFHS is an efficient and effective solution for photovoltaic

cooling applications.

Khanna et al. [81] examined how PCM can be incorporated into fins affixed to solar PV systems to maximize their efficiency. A number of experiments were conducted by the authors to determine how PCM-equipped fins performed under various conditions. PCM-equipped fins can increase the efficiency of PV systems by reducing the temperature of PV cells, thus enabling them to produce more power. By using fins together with PCM, the authors found that thermal losses to the environment can be minimized, thereby increasing efficiency. In experiments, PCM was found to maximize PV system performance when installed on fins.

Using finned solar photovoltaic phase-change materials, Khanna et al. [82] optimized their system's performance. Performance of the system was evaluated by examining fin length, fin number, and fin spacing. Thermal insulation materials are also taken into account when analysing the performance of the system. In order to validate the results of the analytical model, the authors conducted experiments. A finned PV PCM system's performance is significantly affected by the length, number, and spacing of its fins. By incorporating insulation material into finned PV PCM systems, this paper offered valuable information regarding their optimization.

The finned heat sink prototype for "low-concentration photovoltaics (LCPV)" systems was examined by Gallero et al. [83]. In order to assess thermal performance, the authors compared the results of analytical correlations and CFD simulations. Heat sinks with fins were tested under a variety of conditions to determine their heating capacity. Both methods accurately predicted the thermal performance of the heat sink. Thermal performance was predicted more accurately and more reliably using CFD modelling in extreme cases. LCPV systems that use finned heat sinks might benefit from CFD modelling as a method of assessing their thermal performance.

With a passive cooling method using phase change material in a finned container heat sink, Wongwuttanasatian et al. [84] studied how a PV module can be enhanced in terms of performance. A phase change material was added to the PV module and was found to significantly improve its thermal performance. A further 11.2% increase in power output was achieved. According to the authors, this cooling system could increase a photovoltaic system's efficiency and lifetime.

A new model was proposed by Vittorini and Cipollone [85] for illustrating the performance of PV modules in real-world scenarios. Multiple thermal and electrical parameters, such as temperature and efficiency, were accounted for in the model to predict a module's performance accurately. Based on the model's accuracy in predicting the performance of a fin-cooled photovoltaic module under a variety of conditions, the authors tested it on a fin-cooled module. As a result of the model, PV modules will be more efficient and solar energy technology will progress further.

According to Luo et al. [86], "flared-fin heat sinks" for concentrator photovoltaics (CPVs) perform well at dissipating heat. CPV modules were designed with different materials, shapes, and numbers of fins, and four different module designs were tested. Heat sinks with flared fins have been shown to successfully disperse heat, and they are dependent on the intended thermal capacity to determine their best design. In the authors' conclusions, flared-fin heat sinks are a viable option for CPV modules, although further research is needed to improve their design.

In a study by Kandeal et al. [87], the effects of climate on finned PCM solar panels were investigated. A series of experiments was conducted by the authors to test how well the system performed when exposed to different weather conditions. According to the data, the combination of finned PCMs and photovoltaic systems stabilizes the system's temperatures and enhances its electrical output even in humid and hot conditions. Moreover, placing the finned PCM between the solar panel and battery was found to enhance its efficiency. It appears that the finned PCM integrated PV system would be suitable for use in hot, humid environments. Modern photovoltaic cooling methods have been extensively reviewed, categorized, and discussed by the authors. Radial cooling, water and air cooling with or without fins, and phase change material cooling were all reviewed. This article, however, focuses on evaluating the use of fins in variable PV/T systems.

In extreme heat conditions, Kumar et al. [88] examined the performance of a PV module integrating a PCM filled container and external fins. Testing was carried out at temperatures ranging from 20 °C to 50 °C on the PV-PCM system. Compared with a PV-PCM system without external fins, the PV-PCM system with fins performed significantly better. The PV panel backside temperature was reduced by over 5.8° Celsius, and the system's efficiency increased by 11.3%. It can be concluded that the PV-PCM system with external fins is an effective solution for improving PV energy efficiency in hot climates.

According to Johnston et al. [89], silicon PV cells were studied for the effectiveness of a finned heat sink in cooling, and the effects of inclination angles on productivity. In an array of finned heat sinks that have varying inclination angles, the authors simulated a flat-plate silicon photovoltaic cell using a finite element model and a thermal network. A 45° angle resulted in the greatest efficiency of the photovoltaic cell, with inclination angles between 0° and 90° significantly reducing peak temperature. Researchers found that using finned heat sinks at an angle may be a useful method for reducing temperatures and increasing photovoltaic efficiency.

In a study conducted by Shoeibi et al. [90], finned PV/T solar air dryers with compound parabolic concentrators (CPCs) were investigated. By optimizing the dryer's design parameters, the authors analyzed the operational performance of the dryer. In the CPC, the PV/T finned system increased the dryer's efficiency and drying rate. Aside from being stable and reliable, the system also performed well. According to the results of this study, finned PVT solar air dryers with CPCs may prove to be economical and efficient methods for drying.

An experimental direct-expansion solar-assisted heat pump was developed by Song et al. [91] using a "hybrid compound parabolic concentrator/photovoltaic/fin evaporator". System performance was evaluated under a variety of conditions, such as varying temperatures and sunlight intensities. It was found that the system reached a COP of more than 4.5 at various settings, with a maximum of 6.2. In addition, a maximum of 16.1 kW of power was generated by the system. Researchers concluded that installing a solar-assisted heat pump system could result in significant reductions in energy consumption and costs.

An HCPV/T hybrid with a nanofluid-cooled pin-finned heatsink was studied by Dey et al. [92]. Researchers used "computational fluid dynamics (CFD)" to analyze the thermal performance of heatsinks based on various parameters, such as fin number, fin height, and arrangement. In comparison with flat-finned heatsinks, pin-finned heatsinks achieved better thermal performance. Also, the

pin-finned heatsink performed better in terms of exergy efficiency, exergy destruction, and heat transfer rate than the flat-finned heatsink. HCPV/T hybrid systems are better suited to pin-finned heatsinks than flat-finned ones, conclude the authors.

According to Bassam et al. [93], nano PCM and micro-fin tubes filled with nanofluid work effectively as a photovoltaic thermal collector. PVT system thermal characteristics were measured experimentally by the researchers. Thermal performance was improved

Table 1
Comparison of passive cooling.

References	Fin Type	Cooling temperature T (°C)	Dimensions	Design/System	Focus/Key findings
[78]	wavy channels	Cooling T of up to 15 °C	width of 0.4 mm, a pitch of 3 mm, and a height of 0.5 mm	Micro-fins that are designed to enhance the efficiency of passive concentrating photovoltaic (PV) cooling through natural convection are used.	Research has uncovered that the utilization of plate micro-fins can boost the rate of thermal transfer and lower the temperature of the photovoltaic component.
[79]	Aluminium finned plate	Reduce by −5 °C, Cooling T of up to 20 °C	base of the fin (20 mm) and the height of the fin (5 mm)	"finned plate of aluminium to improve PV panel"	"improving PV panel performance using a finned plate of aluminium"
[80]	trapezoidal channel	Cooling to 20–45 °C & lowest cooling T is 65.4 °C	2 mm in height and 4 mm in width	truncated multi-level fin heat sink	focus on photovoltaic cooling, PV heat dissipation
[81]	Aluminium flat plate	Cooling to 23-35	15 mm in height and 15 mm in width	"optimization of fins fitted PCM equipped solar PV"	
[82]	Aluminium flat plate	Cooling to 15–20 °C	30 mm in height and 15 mm in width	optimization of finned solar PV PCM	focus on the effect of climate and the optimization of the system
[83]	Aluminium flat plate	Reduced by −8 °C, Cooling T of up to 32 °C	base of the fin (20 mm) and the height of the fin (10 mm)	"assessing thermal performance of a finned heat sink prototype"	"assessing the thermal performance of a finned heat sink prototype for Low Concentration PV"
[84]	- Grooved container - Tubed container - Finned container.	Cooling T of up to 37 °C	3 mm in height, 9 mm in length, and 2 mm in thickness	Using PCM to increase the heat transfer efficiency of the fins	"It is found that using PCM in a finned container heat sink was able to increase the performance of a photovoltaic module by enhancing the passive cooling rate."
[85]	Aluminium flat plate	–	–	effects of fin-cooled PV modules on efficiency and performance mapping	It is found that fin-cooled PV module model provided an accurate mapping of the performance of the module and could be used to assess the electric efficiency of the module under real operating conditions
[86]	flared-fin	Cooling T of up to 40 °C	length of 250 mm, a fin width of 6 mm, a pitch of 60 mm, and a height of 8 mm	flared-fin heat sinks that are used to improve the heat dissipation performance of concentration PV modules	"It is found that the heat transfer performance of the flared-fin heat sink was higher than that of the flat-fin heat sink, and that the use of a flared-fin heat sink could increase the heat dissipation performance by up to 16.4%."
[87]	parallel-finned aluminium plate Vertically staggered fins	Cooling to 20–30 °C	20 mm in height and 15 mm in width	"effect of climate on the electrical performance of finned PCM integrated solar PV"	focuses on the optimization of the system
[88]	External fins	Cooling T of up to 30 °C	width of 7 mm, a pitch of 30 mm, and a height of 15 mm	fins are designed to absorb heat from the PV modules and transfer it to the PCM	focus on the effect of climate and the optimization of the system
[89]	Aluminium inclined flat plate	Reduced by −4 °C, Cooling T of up to 40 °C	base of the fin (25 mm) and the height of the fin (6 mm)	"finned heat sinks and the effect of inclination angle"	finned heat sinks to cool silicon PV cell
[90]	V-shape fins	Reduce by −5 °C	0.5 mm in thickness, 15 mm in height, and 41 mm in length	fins are designed to increase the heat transfer rate within the solar air dryer	It is found that the use of a finned PV/T solar air dryer with a compound parabolic concentrator could improve the air temperature and humidity, as well as increase the drying rate.
[91]	finned-tube evaporator	Reduce by −10	0.3 mm in thickness, 8 mm in height, and 60 mm in length	fins are designed to increase the efficiency of the direct-expansion solar-assisted heat pump	It is found that PV/T with fins evaporator "direct-expansion solar-assisted heat pump is capable of achieving greater heating capacity and coefficient of performance" (COP).
[92]	Inline and staggered shaped pin-fins	Cooling to 25–75 & lowest cooling T is 56.3	0.3 mm in height and 0.7 mm in width	PV/T collector "with nano PCM and micro-fins tube nanofluid"	focus on photovoltaic cooling, PV heat dissipation

by adding nano PCM and nanofluid to the system. As a result of the study, the nano PCM and nanofluid are effective in enhancing the utilization of renewable energy through PVT. A comparison of the literature studies in terms of temperature, fin dimensions and design, and key findings can be found in Table 1.

Compared to active cooling systems such as fluid pumps [94], nanofluids [95], heat exchangers [96], etc., passive cooling reduces many complications and costs. In their review of PV passive cooling devices [94], Mahdavi et al. discussed similar topics. PV/T systems were studied in detail in relation to natural ventilation, water immersion, and heat pipes. According to the authors, natural ventilation can be considered the most cost-effective alternative for most problems when it comes to passive cooling. In the current study, several recent research studies were reviewed, focusing on fins (their shapes, types, compositions, arrangements, designs, and materials), along with their benefits and drawbacks.

4.6.5. Arrangement and design of fins

In photovoltaic modules, straight fins are most commonly used to cool them. In this configuration, a series of parallel fins are arranged with air channels running between them to help dissipate heat from the module. A louvered, curved, or twisted fin is also a common fin configuration [97,98].

Photovoltaic fins cooling method design criteria include [99,100]:

1. Heat must be dissipated efficiently by the fins.
2. It is important to design the fins to maximize airflow around the photovoltaic modules.
3. A fin's design must minimize dust and debris accumulation.
4. In order to maximize the module's efficiency, the fins must minimize direct sunlight.
5. Photovoltaic modules must have fins that provide uniform temperatures.
6. It is essential that the fins are lightweight and durable at the same time.

Literature has investigated many types of fins, including:

1. Airflow Fin Arrangement: Used for cooling solar panels using natural air flow [101].
2. Open fin design: Air passes through the fins, cooling PV modules [102].
3. The closed fin design: Allows photovoltaic PV panels to be more efficiently cooled by enclosing them in an enclosed space [103].
4. Designed with louvered fins: Provides greater cooling surface area for photovoltaic panels [104].
5. Design with Canted Fins: Improves heat transfer from PV panels to the atmosphere [105].
6. Design with Curved Fins: Creates a greater flow of air over the panels of a photovoltaic system [106].
7. Designs with pin fins: Enhance cooling efficiency by increasing the surface-to-volume ratio [107].
8. Design with V-Shaped Fins: Provides better cooling than traditional flat fins [108].
9. Design with Extended Fins: Provides greater cooling surface area for photovoltaic panels [109].
10. Design with perforated fins: For increasing airflow over photovoltaic panels [110].

A fin type's efficiency and heat transfer capacity is determined by its shapes and geometries. as follows [111,112]:

1 Delta Wing Fin
2 Flat Plate Fin
3 Extended Surface Fin
4 Pin Fin
5 Louvered Fin
6 Perforated Fin
7 Microchannel Fin
8 Tabulator Fin

The advantages and disadvantages of fin length, thickness, arrangement symmetry, and material properties are:

1 ***Fin Length*** [111]:

Advantages: In addition to providing more surface area, longer fins can also provide more heat transfer efficiency.
Disadvantages: The longer the fin, the heavier, the more expensive, and the more likely it is to be damaged.

2 ***Fin Thickness*** [112]:

Advantages: A thicker fin provides better heat transfer efficiency since more material can absorb and dissipated more heat.
Disadvantages: It may take more space to fit thicker fins and they are more expensive.

3 ***Fin Arrangement Symmetry*** [113]:

Advantages: Heat transfer can be improved by symmetrical fin arrangements.
Disadvantages: Manufacturing symmetrical fin arrangements requires specialized tools.

4 . ***Fin Material Properties*** [114]:

Advantages: High thermal conductivity and low thermal resistance fin materials can provide significantly better performance than materials with low thermal conductivity.

Disadvantages: Material costs may be higher for high-performance fins and manufacturing processes may be more complicated.

Literature has proposed and tested a variety of fin configurations, such as:

4.7. Cross-fin configuration

Cross-fin refers to a specific design configuration commonly used in heat exchangers and heat transfer devices. It involves the arrangement of fins in a crisscross pattern to enhance heat transfer efficiency. In a cross-fin heat exchanger, the fins are typically thin, elongated structures positioned perpendicular to the fluid flow direction. These fins create a large surface area that facilitates the transfer of heat between the fluid flowing through the heat exchanger and the surrounding environment [115,116].

The crisscross pattern of cross-fin heat exchanger fins offers several advantages. First, it increases the effective surface area available for heat transfer, allowing for improved heat exchange between the fluid and the surrounding medium. This increased surface area promotes increased heat dissipation or heat absorption, depending on the specific application [117]. Second, the cross-fin configuration enhances fluid mixing and turbulence within the heat exchanger. As the fluid flows across the fins, it encounters obstructions and irregularities, which disrupt the flow and promote better mixing. This turbulent flow helps break up the boundary layer near the fin surface, increasing the convective heat transfer coefficient and enhancing overall heat transfer efficiency [118].

The cross-fin type is commonly employed in various heat transfer applications, including air-cooled heat exchangers, radiators, and condensers. It is particularly effective in situations where there is a large temperature difference between the fluid and the surrounding environment or when efficient heat dissipation is required. As shown in Fig. 3, the fins create a lattice-like structure that allows air to flow between them, increasing cooling air flow and reducing heat build-up [119].

A cross-fin heat sink was examined by Feng et al. [120]. Researchers found that heat transfer coefficients increased with fin spacing, fin thickness, and inlet temperature, but decreased with fin height as fin spacing, fin thickness, and inlet temperature increased. Moreover, higher Reynolds numbers resulted in higher heat transfer coefficients, the authors found. Based on these results, it is possible to improve the efficiency of a cross-fin heat sink by optimizing its design.

4.8. Perforated-fin base configuration

The perforated-fin base configuration is a specific design used in heat exchangers and heat transfer devices to improve heat transfer efficiency. In this configuration, the fins are perforated, meaning they have small holes or openings in them. This allows for enhanced heat exchange between the fluid and the surrounding environment. Perforated-fin base design increases heat transfer surface area while maintaining structural integrity [121]. By introducing perforations in the fins, the effective surface area for heat exchange is significantly expanded, resulting in improved heat transfer performance. The fin perforations serve multiple functions. Firstly, they enhance fluid mixing and turbulence. As the fluid flows through the perforated fins, it encounters obstructions and irregularities, causing turbulence and better mixing. This turbulent flow disrupts the boundary layer near the fin surface, increasing the convective heat transfer coefficient and facilitating more efficient heat transfer [122]. Secondly, the perforations allow for better airflow or fluid flow distribution across the heat exchanger surface. They help to evenly distribute the fluid or air over the entire fin surface area, maximizing the utilization of the heat transfer surface. This ensures uniform heat transfer across the device [123].

The specific configuration of perforations, such as their size, shape, and distribution, can be optimized based on the requirements of the particular application [124]. These parameters can affect factors like pressure drop, heat transfer performance, and structural integrity, and they need to be carefully designed and balanced. The perforated-fin base configuration finds application in various heat transfer systems, including air-cooled heat exchangers, heat sinks for electronics, and condensers. It is particularly useful in situations where efficient heat dissipation is needed or where a high surface area for heat transfer is desired, as shown in Fig. 4.

Using an array of horizontal fins, Huang et al. [125] examined how holes in the bottom affected convection-based heat transfer. In order to measure the effectiveness of heat transfer, different perforation sizes and fin counts were tested. Researchers found that adding perforations to the fin base improved natural convection heat transfer. Fin number and perforation size are both factors that affect the heat transfer rate the most.



Fig. 3. Cross-fin configuration [119].

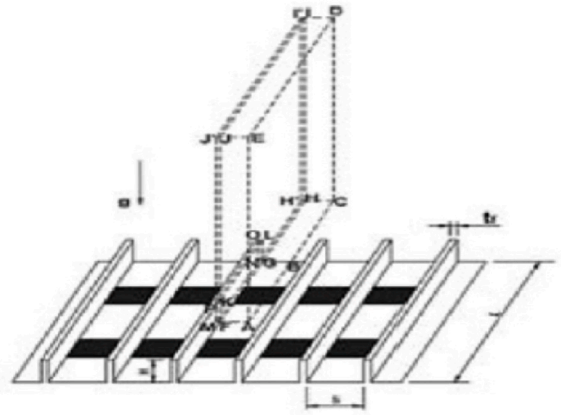


Fig. 4. Perforated fin configuration [124].

4.9. Lateral perforated-fin configuration

Lateral perforated-fin is a cooling technology commonly used in electronic devices to dissipate heat and prevent overheating. It involves the use of perforated metal fins that are arranged in a lateral orientation, typically attached to heat sinks or heat exchangers [126]. The main purpose of lateral perforated-fin is to enhance heat transfer efficiency by increasing the surface area available for heat dissipation. The perforations in the fins allow for improved airflow and better distribution of heat across the fins. This design enables effective cooling of electronic components, such as processors, graphics cards, and power modules, which generate significant amounts of heat during operation [127].

The benefits of lateral perforated-fin include [128,129]:

Efficient heat dissipation: The increased surface area provided by the perforated fins allows for faster and more efficient heat transfer. This helps in maintaining optimal operating temperatures for electronic devices, thereby enhancing their performance and reliability.

Enhanced airflow: The perforations in the fins facilitate improved airflow, allowing for better ventilation and heat distribution. This prevents hot spots and ensures more uniform cooling throughout the device.

Compact design: Lateral perforated-fin cooling systems can be designed to be relatively compact, making them suitable for applications with limited space, such as laptops, gaming consoles, and compact electronic devices.

Cost-effective: Compared to some alternative cooling methods, lateral perforated-fin cooling can be a cost-effective solution, as it relies on relatively simple manufacturing techniques and materials.

However, there are also some disadvantages associated with lateral perforated-fin cooling [130,131]:

Pressure drop: The presence of perforations in the fins can lead to increased pressure drop, causing a slight reduction in overall system efficiency.

Limited fin density: The arrangement of lateral fins may limit the number of fins that can be used, which can impact the total surface area available for heat dissipation.

Noise generation: In certain cases, the design of lateral perforated-fin cooling systems may lead to increased noise levels due to the interaction between the airflow and the perforated fins.

According to Shaeri and Bonner [132], the laterally perforated fins on a heat sink contribute to heat transfer, as shown in Fig. 5. The study examines the convective heat transfer coefficient as a function of fin geometry, heat sink length, and perforation width. As the heat sink length and perforation width increased, the heat transfer coefficient increased. Moreover, the heat sink with laterally

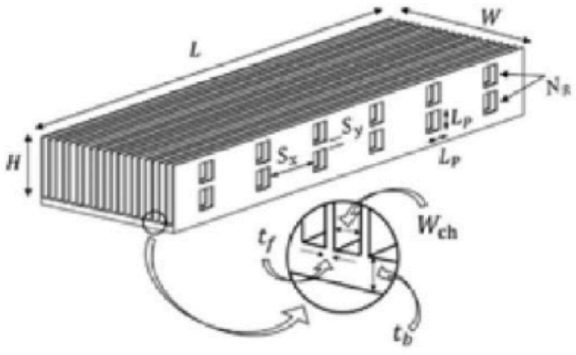


Fig. 5. Lateral perforated-fin configuration [132].

perforated fins performed better. Enhanced heat transfer can be achieved by designing heat sinks with laterally perforated fins.

4.10. Displaced fin configuration

In this process, a finned heat sink is used, a heat conducting device composed of many thin, closely spaced metal fins. Air flows easily over these fins, dissipating heat away from the photovoltaic module due to their large surface area. Abbas and Wang [97] show how displacement design can enhance natural convection heat sinks in Fig. 6. Researchers analyzed displacement design's effects on natural convection heat sinks in an experimental study. Displacement design contributed significantly to convective heat transfer coefficient increases as a result of the study. Moreover, the authors suggested that further research into the topic would benefit from displacement design.

4.11. Triangular fin configuration

Triangular fin configuration is a type of cooling design that involves the use of triangular-shaped fins for heat dissipation. The fins are arranged in a triangular pattern, typically attached to heat sinks or heat exchangers. This configuration is often employed in various electronic devices, such as computer processors, power amplifiers, and LED lights, where efficient cooling is essential to prevent overheating and maintain optimal operating temperatures [133].

The benefits of a triangular fin configuration include [134,135]:

Increased surface area: The triangular shape of the fins provides a larger surface area compared to traditional rectangular or cylindrical fins. This enhanced surface area allows for better heat transfer and improved cooling performance.

Enhanced airflow: The triangular configuration promotes better airflow between the fins, improving the overall ventilation and heat distribution within the cooling system. This helps to prevent hot spots and maintain more uniform cooling across the device.

Compact design: The triangular fin configuration can be designed to be more compact, allowing for efficient heat dissipation in devices with limited space constraints.

Enhanced structural stability: The triangular shape offers improved structural stability and rigidity to the fins, making them less prone to bending or deformation during operation or handling.

However, there are also some potential disadvantages associated with the triangular fin configuration [136,137]:

Manufacturing complexity: The production of triangular fins may involve more intricate manufacturing processes compared to traditional rectangular or cylindrical fins, potentially increasing production costs.

Restricted airflow direction: The triangular shape of the fins may limit the direction of airflow through the cooling system, potentially requiring careful design considerations for optimal cooling efficiency.

Specific application requirements: The effectiveness of the triangular fin configuration may vary depending on the specific application and cooling requirements. It may not be suitable for all types of electronic devices or heat dissipation scenarios. It's worth noting that the benefits and disadvantages mentioned above are general observations and may vary depending on the specific implementation and design considerations.

Using an alternating arrangement, Chu et al. [138] assessed the efficiency of a passive heat sink. Thermal performance was evaluated experimentally and numerically as shown in Fig. 7. By reducing the maximum heat sink temperature by up to 11.4%, the alternating layout was found to be more effective than the traditional layout. A maximum heat dissipation rate of 7.3% was achieved with the alternating layout, which was also better than the traditional layout.

4.12. Wavy fin configuration

A wavy fin configuration is a cooling design that involves the use of fin structures with a wavy or undulating shape (Fig. 8). The fins are typically made of metal or other thermally conductive materials and are used for heat dissipation in various applications. Wavy fin

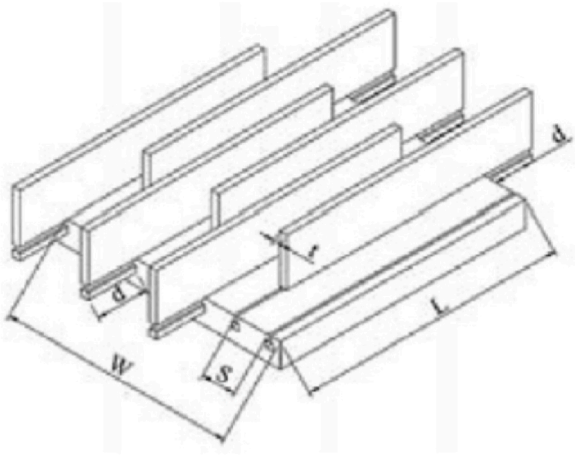


Fig. 6. Displaced fin configuration [97].

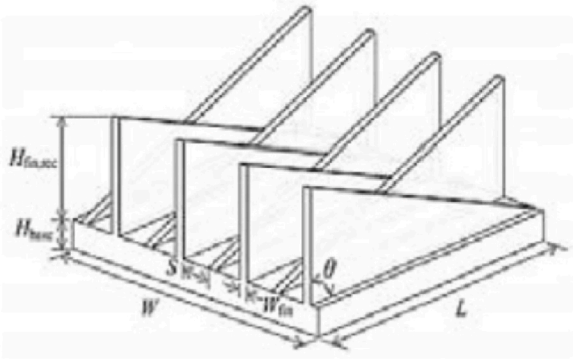


Fig. 7. Triangular fin configuration [138].

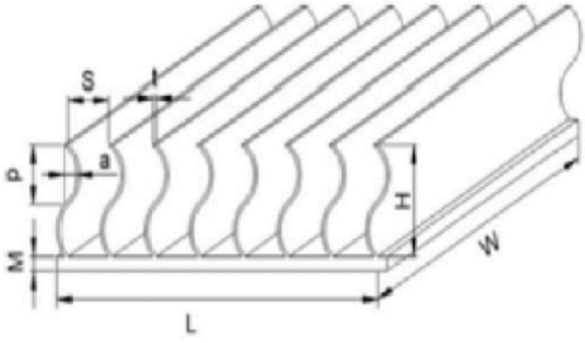


Fig. 8. Wavy fin configuration [139].

configurations are commonly employed in heat sinks and heat exchangers, particularly in electronic devices where efficient cooling is crucial. They can be found in computer processors, power electronics, LED lights, and other devices that generate substantial amounts of heat during operation [139].

The benefits of a wavy fin configuration include [140,141]:

Increased surface area: The wavy shape of the fins provides a larger surface area compared to straight or flat fins. This increased surface area allows for better heat transfer and improved cooling efficiency.

Enhanced turbulence and heat transfer: The undulating shape of the fins disrupts the airflow, creating turbulence. This turbulence enhances the heat transfer process by facilitating better mixing and improving heat distribution across the cooling system.

Improved airflow direction: The wavy fin configuration can be designed to direct airflow in a specific pattern, optimizing the cooling performance. It can help in channelling the air through desired paths, maximizing heat dissipation from critical components.

Structural rigidity: The wavy shape adds structural rigidity to the fins, making them less prone to bending or deformation during operation or handling.

However, there are also some potential disadvantages associated with wavy fin configurations [142,143]:

Manufacturing complexity: The production of wavy fins may involve more intricate manufacturing processes compared to straight fins, potentially increasing production costs.

Increased pressure drop: The undulating shape can create higher resistance to airflow, leading to increased pressure drop across the cooling system. This may slightly reduce overall system efficiency [144].

Limited applicability: The effectiveness of wavy fin configurations may vary depending on the specific application and cooling requirements. It may not be suitable for all types of devices or heat dissipation scenarios.

It's important to note that the benefits and disadvantages mentioned above are general observations and may vary depending on the specific implementation and design considerations [145]. The most important studies on plate fins-tube have been collected and compiled by Macchitella et al. [145]. Depending on the size or gauge, these pipes are also classified as small, large, or multi-gauge. Different approaches to developing tube designs are described by the authors. A discussion of possible future developments and research areas was also included in the paper.

An increase in heat sink performance using a displacement design was examined by Abbas and Wang [97]. By using computational fluid dynamics simulations, the authors found that displacement designs can improve heat transfer performance up to 22.9%. In this study, we demonstrate that displacement design can improve heat transfer performance in a viable and effective manner.

4.13. Truncated multi-level fin configuration

Truncated multi-level fin configuration is a cooling design that involves the use of multi-level fins with truncated or stepped shapes. This configuration is commonly employed in heat sinks and heat exchangers to enhance heat dissipation and improve cooling efficiency [146]. In this design, the fins are arranged in multiple levels or stages, with each level having a different height or length. The fins are typically made of thermally conductive materials such as metal or aluminium. The purpose of this configuration is to increase the surface area available for heat transfer and promote better airflow within the cooling system. Truncated multi-level fin configurations are widely used in various applications, including computer processors, power electronics, LED lights, and other devices where efficient cooling is critical [147].

The benefits of a truncated multi-level fin configuration include [148,149]:

Increased surface area: The multi-level arrangement provides a larger surface area compared to single-level fin designs. This increased surface area allows for better heat dissipation and improved cooling performance.

Enhanced airflow and heat transfer: The stepped or truncated shape of the fins helps to create turbulence in the airflow, promoting better mixing and heat distribution. This results in improved heat transfer efficiency and more effective cooling of the components.

Structural stability: The multi-level design with stepped fins offers enhanced structural stability and rigidity, reducing the risk of bending or deformation during operation or handling.

Compact design: Despite having multiple levels of fins, the truncated configuration can be designed to be relatively compact, making it suitable for applications with space constraints.

However, there are also potential disadvantages associated with truncated multi-level fin configurations [150,151]:

Manufacturing complexity: The production of truncated multi-level fins may involve more intricate manufacturing processes compared to simpler fin designs, potentially increasing production costs.

Pressure drop: The stepped shape and multi-level arrangement can result in increased resistance to airflow, leading to higher pressure drop across the cooling system. This may slightly reduce overall system efficiency.

Limited applicability: The effectiveness of truncated multi-level fin configurations may vary depending on the specific application and cooling requirements. It may not be suitable for all types of devices or heat dissipation scenarios.

It's important to note that the benefits and disadvantages mentioned above are general observations and may vary depending on the specific implementation and design considerations.

In a paper [30], Al-Waeli et al. examined the effect of a truncated multilevel fin heat sink on heat dissipation during natural convection during photovoltaic cooling, as illustrated in Fig. 9. MLFHS with truncated fins was found by the authors to have higher heat dissipation and reduced thermal resistance. Additionally, the authors discovered that MLFHS with truncated fins dissipate more heat than those without, and that the thermal resistance would decrease with increasing fin count. For photovoltaic cooling applications, this research provides insight into heat sink design optimization.

4.14. Cut-corner plate fin configuration

Cut-corner plate fin configuration is a cooling design that involves the use of plate-like fins with cut corners. This configuration is commonly employed in heat sinks and heat exchangers to enhance heat dissipation and improve cooling efficiency. In this design, the fins are typically made of thermally conductive materials such as aluminum and are attached to a base plate. The corners of the fins are cut at angles, creating a cut-corner shape. This design allows for increased surface area and improved airflow within the cooling system. Cut-corner plate fin configurations are used in various applications, including electronic devices such as computer processors, power electronics, and LED lights, where efficient cooling is essential [152].

The benefits of a cut-corner plate fin configuration include [153,154]:

Increased surface area: The cut-corner shape of the fins provides a larger surface area compared to traditional rectangular or flat plate fins. This increased surface area allows for better heat dissipation and improved cooling performance.

Enhanced airflow and heat transfer: The cut-corner shape promotes better airflow between the fins, reducing air resistance and increasing heat transfer efficiency. This helps in maintaining optimal operating temperatures and preventing overheating of the

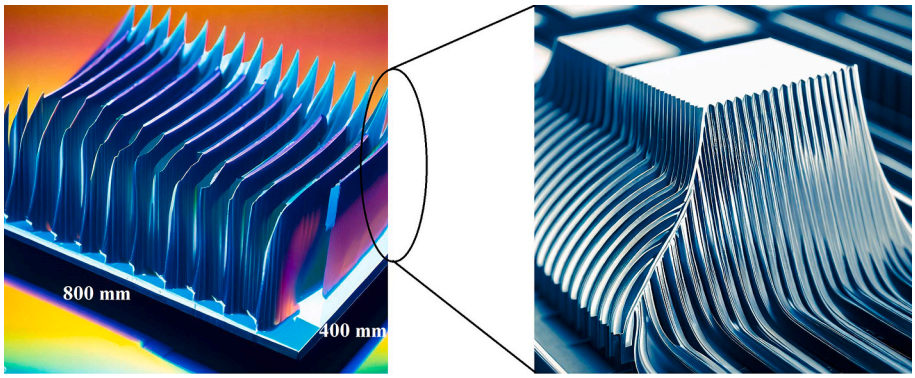


Fig. 9. Truncated multi-level fin configuration [151].

components.

Structural stability: The plate-like design with cut corners offers structural stability and rigidity to the fins, making them less prone to bending or deformation during operation or handling.

Compact design: Despite the increased surface area, the cut-corner plate fin configuration can be designed to be relatively compact, making it suitable for applications with space constraints.

However, there are potential disadvantages associated with cut-corner plate fin configurations [155,156]:

Manufacturing complexity: The production of cut-corner plate fins may involve more intricate manufacturing processes compared to simpler fin designs, potentially increasing production costs.

Pressure drop: The irregular shape of the cut-corner fins can lead to increased pressure drop, causing a slight reduction in overall system efficiency.

Limited heat transfer direction: The cut-corner shape may influence the direction of heat transfer and airflow, requiring careful design considerations for optimal cooling efficiency in specific applications.

It's important to note that the benefits and disadvantages mentioned above are general observations and may vary depending on the specific implementation and design considerations.

As shown in Fig. 10, Meng et al. [157] explored the natural convection heat transfer of a straight-fin heat sink. A numerical model was constructed based on the data collected during testing of the heat sink. A satisfactory agreement was found between the model's results and the experimental data. The results of the study showed that the fin spacing, fin length, and the direction of the air flow greatly affect natural convection heat transfer of straight-fin heat sinks. For optimum performance, fin spacing should not be too large for optimal heat transfer coefficient, but fin length and spacing should be increased to increase it.

4.15. Inverted trapezoid fin configuration

Inverted trapezoid fin configuration refers to a specific design of fins used in photovoltaic/thermal (PVT) systems [158].

Advantages of using an inverted trapezoid fin configuration in PVT systems include [159,160]:

Enhanced heat transfers: The design of inverted trapezoid fins allows for improved heat transfer efficiency. The shape creates a larger surface area, increasing the contact area between the fin and the surrounding air or fluid, thereby enhancing heat dissipation.

Reduced thermal resistance: The larger surface area provided by the inverted trapezoid fins enables more effective heat dissipation, leading to reduced thermal resistance. This results in lower operating temperatures for the PV cells, which can enhance their electrical efficiency and prolong their lifespan.

Compact design: The inverted trapezoid fin configuration allows for a more compact design of PVT systems. The fins can be arranged in a way that maximizes the heat transfer surface area within a given space, making it suitable for applications with limited available area.

Disadvantages of using an inverted trapezoid fin configuration in PVT systems include [161,162]:

Manufacturing complexity: The design of inverted trapezoid fins may require more complex manufacturing processes compared to traditional fin configurations. This can potentially increase production costs and pose challenges in mass production.

Potential for dirt accumulation: The unique shape of the inverted trapezoid fins may create pockets or crevices where dust, dirt, or other debris can accumulate. This can hinder heat transfer and reduce the overall performance of the PVT system. Regular cleaning and maintenance may be required to mitigate this issue.

Limited compatibility: Inverted trapezoid fin configurations may not be compatible with all PVT system designs. The specific requirements of a particular system, such as space constraints or integration with other components, may limit the feasibility or effectiveness of using this fin configuration.

It is important to note that the advantages and disadvantages mentioned above are general considerations and may vary depending

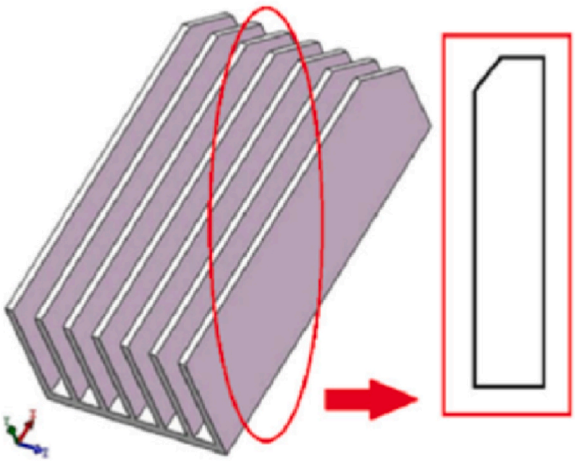


Fig. 10. Cut-corner plate fin configuration [157].

on the specific application and implementation of the inverted trapezoid fin configuration in a PVT system.

A novel heat dissipation fin design is presented by Charles and Wang [163] for enhancing natural convection, as shown in Fig. 11. To evaluate the performance of the fins, the authors conducted an experiment. Approximately 8–10% more convective heat was transferred using the novel fin design. As a result of the novel design, the authors suggest that it is useful for applications requiring heat dissipation utilizing natural convection. This paper found that compared to traditional designs, the Inverted *trapezoid fin* design increased convective heat transfer by 8–10%, ensuring its viability for applications requiring natural convection.

4.16. Dual-height plate-fins configuration

Dual-height plate-fins configuration refers to a design of heat transfer fins used in photovoltaic/thermal (PVT) systems [164].

Advantages of using a dual-height plate-fins configuration in PVT systems include [165,166]:

Improved heat transfer: The dual-height design of plate-fins creates multiple channels for the flow of air or fluid, enhancing the heat transfer process. The fins provide increased surface area and promote turbulence, resulting in efficient heat dissipation from the PV cells. This improves the overall thermal performance of the system.

Enhanced airflow: The dual-height configuration enables better airflow distribution over the PV cells. The varying heights of the plate-fins create a non-uniform pressure distribution, which helps to evenly distribute the airflow across the entire surface of the PV panels. This reduces hot spots and enhances the system's cooling effectiveness.

Flexible design: The dual-height plate-fins configuration offers flexibility in design and customization. The height variations can be adjusted based on specific system requirements, such as available space, desired thermal performance, and integration with other system components. This allows for optimized heat transfer and improved system efficiency.

Disadvantages of using a dual-height plate-fins configuration in PVT systems include [167,168]:

Increased pressure drop: The presence of multiple channels and varying heights in the plate-fins design can lead to increased pressure drop across the system. This may require higher fan power or fluid pumping power to maintain the desired airflow or fluid flow rate, which can result in increased energy consumption.

Complex manufacturing and assembly: The design of dual-height plate-fins can involve more complex manufacturing processes and assembly requirements compared to simpler fin configurations. This can lead to higher production costs and potential challenges in mass production.

Maintenance and cleaning difficulties: The presence of multiple channels and varying heights in the plate-fins configuration can make cleaning and maintenance more challenging. Dust, debris, or contaminants may accumulate in the smaller gaps and crevices, requiring careful and thorough cleaning procedures to maintain optimal heat transfer performance.

It is important to note that the advantages and disadvantages mentioned above are general considerations and may vary depending on the specific application and implementation of the dual-height plate-fins configuration in a PVT system.

As shown in Fig. 12, Jeon and Byon [169] studied the "thermal performance of dual-height fin heat sinks subjected to natural convection." A numerical simulation was conducted to evaluate how the number of fins, fin height, and fin thickness affect heat transfer. Heat transfer performance improved with an increase in fin number. Thermal performance was also improved with higher fin heights and thicknesses, but this effect was less pronounced. A plate fin heat sink with two different fin heights is more efficient at cooling than a single-height fin heat sink on the basis of the research.

4.17. Reverse-trapezoidal plate-fins configuration

Reverse-trapezoidal plate-fins configuration refers to a specific design of heat transfer fins used in photovoltaic/thermal (PVT)

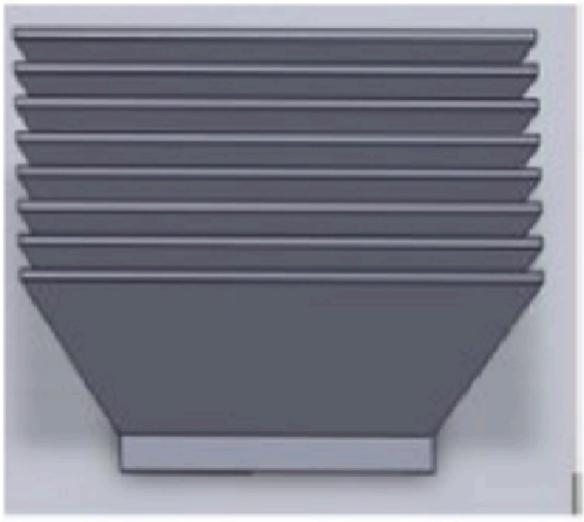


Fig. 11. Inverted trapezoid fin configuration [163].

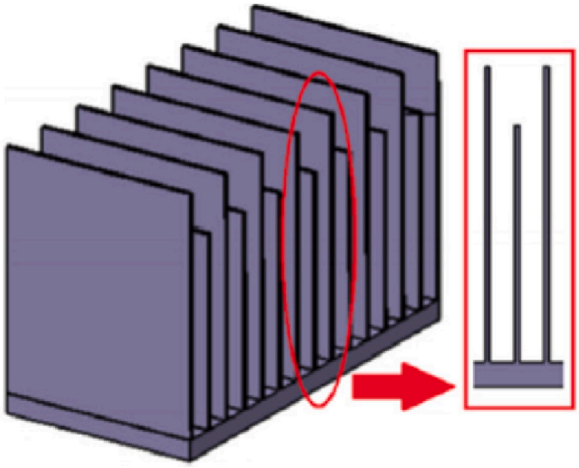


Fig. 12. Dual-height plate-fins configuration [169].

systems [170].

Advantages of using a reverse-trapezoidal plate-fins configuration in PVT systems include [171,172]:

Enhanced heat transfer: The reverse-trapezoidal shape of the plate-fins provides a larger surface area compared to traditional rectangular or parallel designs. This increased surface area promotes better heat transfer between the PV cells and the surrounding air or fluid, resulting in improved thermal performance and higher overall system efficiency.

Improved airflow distribution: The unique design of reverse-trapezoidal plate-fins facilitates improved airflow distribution over the PV cells. The shape helps to evenly distribute the airflow, reducing hot spots and ensuring more uniform cooling across the entire surface area of the panels. This aids in maintaining optimal operating temperatures and maximizing electricity generation.

Space efficiency: The reverse-trapezoidal plate-fins configuration allows for efficient utilization of space in PVT systems. The fins can be arranged in a way that optimizes the available area, providing a larger heat transfer surface within a given space. This is particularly beneficial for installations with limited space or architectural constraints.

Disadvantages of using a reverse-trapezoidal plate-fins configuration in PVT systems include [171,172]:

Manufacturing complexity: The design of reverse-trapezoidal plate-fins may involve more complex manufacturing processes compared to simpler fin configurations. The intricate shape and varying dimensions of the fins can require precise manufacturing techniques, leading to increased production costs and potential challenges in mass production.

Potential for debris accumulation: The reverse-trapezoidal shape of the plate-fins may create small crevices or pockets where dust, dirt, or other debris can accumulate. This accumulation can hinder heat transfer efficiency and reduce the performance of the PVT system. Regular cleaning and maintenance may be necessary to prevent any negative impact.

Limited compatibility: The specific design requirements and constraints of a PVT system may limit the feasibility or effectiveness of using reverse-trapezoidal plate-fins. Factors such as available space, integration with other system components, and desired performance targets may influence the suitability of this fin configuration for a particular application.

It is important to note that the advantages and disadvantages mentioned above are general considerations and may vary depending

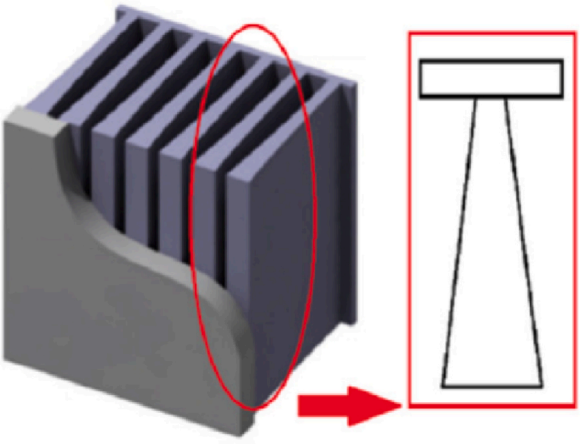


Fig. 13. Reverse-trapezoidal plate-fins configuration [173].

on the specific application and implementation of the reverse-trapezoidal plate-fins configuration in a PVT system.

According to Fig. 13, Kim [173] examined "thermal optimization of plate-fin heat sinks with fins of variable thickness under natural convection". Mathematical analysis was used to determine the effect of fin thickness and spacing on plate-fin heat sink thermal performance. Based on the heat transfer coefficient and the thermal conductivity of the fin material, the optimal fin thickness and spacing were determined. Fin spacing could also be varied to optimize fin thickness. Increasing the fin thickness improved the heat sink's thermal performance, according to the study. Also, the results demonstrated that the fin spacing and thickness of the heat sink affect the thermal resistance, and that it is possible to minimize the thermal resistance by increasing the fin thickness and decreasing the fin spacing. By adjusting the fin thickness and spacing, this paper provides insight into how plate-fin heat sinks can be optimized for thermal performance.

4.18. Plate-fin with dimple configuration

Plate-fin with dimple configuration (Fig. 14) refers to a specific design of heat transfer fins used in PVT systems [174].

Advantages of using a plate-fin with dimple configuration in PVT systems include [175,176]:

Enhanced heat transfer: The dimple configuration on the plate-fins increases the surface area and disrupts the boundary layer, improving heat transfer efficiency. The dimples create turbulence in the airflow or fluid flow, enhancing convective heat transfer and resulting in better cooling of the PV cells. This leads to improved thermal performance and increased system efficiency.

Reduction of pressure drop: The presence of dimples on the plate-fins helps reduce the pressure drop across the system. The turbulence created by the dimples disrupts the flow, minimizing the resistance encountered by the air or fluid passing through the fins. This can lead to lower energy consumption requirements for fans or pumps, contributing to overall energy savings.

Prevention of hot spots: The dimple configuration promotes more uniform airflow or fluid distribution over the PV cells, reducing the likelihood of hot spots. Hot spots can negatively impact the electrical efficiency and lifespan of the PV cells. The dimples help to distribute the heat more evenly, maintaining optimal operating temperatures and maximizing electricity generation.

Disadvantages of using a plate-fin with dimple configuration in PVT systems include [177,178]:

Manufacturing complexity: The design of plate-fins with dimples may involve more complex manufacturing processes compared to simpler fin configurations. The creation of precise and consistent dimples on the fins can require specialized techniques, which may increase production costs and pose challenges in mass production.

Maintenance and cleaning difficulties: The presence of dimples on the plate-fins can make cleaning and maintenance more challenging. Dust, debris, or contaminants may accumulate in the recessed areas, requiring careful cleaning procedures to ensure optimal heat transfer performance. Regular maintenance may be necessary to prevent any reduction in system efficiency due to clogged or dirty dimples.

Potential for flow maldistribution: The presence of dimples can influence the flow distribution over the plate-fins. If not properly designed or implemented, the dimples may cause flow maldistribution, leading to uneven cooling of the PV cells. Careful consideration and optimization of the dimple design are necessary to ensure uniform heat transfer and minimize any negative impact on system performance.

It is important to note that the advantages and disadvantages mentioned above are general considerations and may vary depending on the specific application and implementation of the plate-fin with dimple configuration in a PVT system.

An array of dimple fins was investigated by Chang et al. [179] to observe its effects on natural convection heat transfer. Convective heat transfer was determined by varying the geometric parameters of the fin array mounted on a vertical wall. By incorporating a dimpled fin array, heat transfer through natural convection was increased. It was possible to achieve optimal performance with an

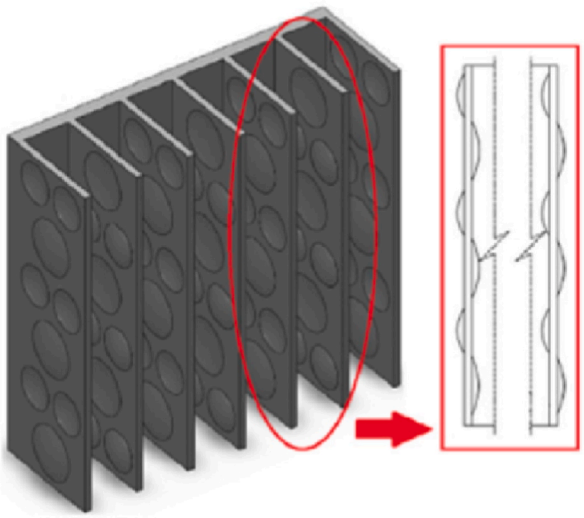


Fig. 14. Plate-fin with dimple configuration [174].

array of greater height, width, and pitch. A fin array requiring improved heat transfer is designed based on the results of this study.

4.19. Cross-fin configuration

Cross-fin configuration refers to a specific design of heat transfer fins used in photovoltaic/thermal (PVT) systems [180].

Advantages of using a cross-fin configuration in PVT systems include [181,182]:

Increased surface area: The cross-fin configuration provides a larger surface area compared to other fin designs. The intersecting fins create additional channels for heat transfer, maximizing the contact area between the PV cells and the surrounding air or fluid. This enhances heat dissipation and improves the overall thermal performance of the PVT system.

Improved airflow distribution: The intersecting fins of the cross-fin configuration help to distribute the airflow more evenly across the surface of the PV cells. This reduces the occurrence of hot spots and ensures more uniform cooling, optimizing the performance and longevity of the PV cells.

Space efficiency: Cross-fin configurations can be designed to make efficient use of available space in PVT systems. The fins can be arranged in a way that maximizes the heat transfer surface area within the given space constraints. This makes cross-fin configurations suitable for installations with limited space or architectural restrictions.

Disadvantages of using a cross-fin configuration in PVT systems include [183,184]:

Increased pressure drop: The complex geometry of the cross-fin configuration can lead to increased pressure drop across the system. The intersecting fins create more resistance to the airflow or fluid flow, resulting in higher energy consumption requirements for fans or pumps to maintain the desired flow rate.

Manufacturing complexity: The design and manufacturing of cross-fin configurations can be more complex compared to simpler fin configurations. The intersecting fins require precise alignment and assembly, which may increase production costs and pose challenges in mass production.

Potential for fouling or blockage: The cross-fin configuration may be susceptible to fouling or blockage due to dust, debris, or other contaminants. The intersections and narrow gaps between the fins can trap particles, potentially reducing heat transfer efficiency and necessitating regular cleaning and maintenance to prevent performance degradation.

It is important to note that the advantages and disadvantages mentioned above are general considerations and may vary depending on the specific application and implementation of the cross-fin configuration in a PVT as shown in Fig. 15, Feng et al. [120] examined how cross-fin geometry affected natural convection in a heat sink. In this study, the authors compared the heat transfer performance of a rectangular cross-fin heat sink and a trapezoidal cross-fin heat sink with a series of experiments and numerical simulations. Compared to rectangular cross-fins, trapezoidal cross-fins performed better at heat transfer. Furthermore, the number of fins and the air velocity of the heat sink improved heat transfer performance. Natural convection in a heat sink is better achieved with trapezoidal cross fins than rectangular cross fins, according to a study.

4.20. Foam -metal fin configuration

Foam-metal fin configuration (Fig. 16) refers to a specific design where metal fins are combined with a foam material in PVT systems [185].

Advantages of using a foam-metal fin configuration in PVT systems include [186,187]:

Enhanced heat transfers: The combination of metal fins and foam material creates a unique configuration that promotes improved heat transfer. The metal fins provide structural support and increase the surface area for heat dissipation, while the foam material enhances convective heat transfer by increasing the contact area between the fins and the surrounding air or fluid. This results in enhanced thermal performance and higher system efficiency.

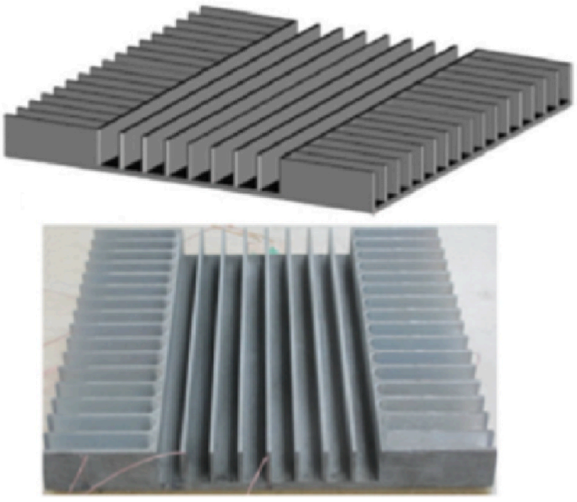


Fig. 15. Cross-fin configuration [120].

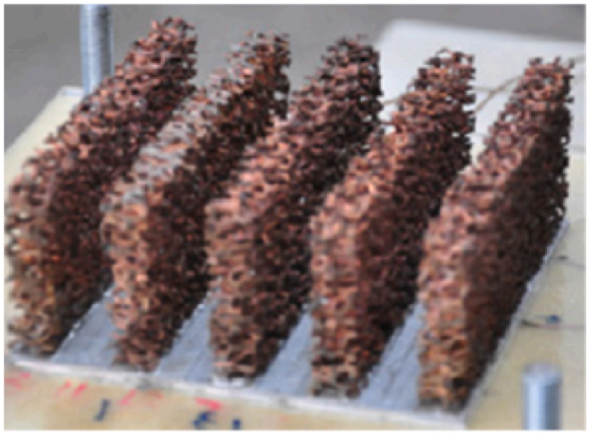


Fig. 16. Foam -metal fin configuration [185].

Lightweight design: The foam material used in the fin configuration is lightweight, which reduces the overall weight of the system. This can be advantageous for installations where weight is a concern, such as rooftop installations or portable PVT systems. The lightweight design facilitates easier transportation, installation, and maintenance.

Insulation properties: The foam material in the fin configuration can provide insulation properties, reducing heat losses from the backside of the PV panels. This helps to maintain higher operating temperatures on the front side, improving the electrical efficiency of the PV cells. The insulation properties of the foam can also help in reducing temperature fluctuations and increasing the stability of the system.

Disadvantages of using a foam-metal fin configuration in PVT systems include [188,189]:

Reduced structural integrity: While the foam material contributes to lightweight design, it may have lower structural integrity compared to solid metal fins. The foam may be more prone to damage or degradation over time, which can affect the overall durability and lifespan of the fin configuration. Careful consideration and selection of robust foam materials are necessary to mitigate this potential issue.

Limited heat dissipation: The foam material may have limitations in its ability to dissipate heat compared to solid metal fins. The foam may have a lower thermal conductivity, which can restrict the heat transfer process. This can lead to increased operating temperatures of the PV cells and potentially impact their electrical efficiency.

Compatibility with manufacturing processes: The incorporation of foam material into the fin configuration may introduce additional complexities in the manufacturing process. Specialized techniques and equipment may be required to ensure proper adhesion and alignment of the foam and metal components. This can result in increased manufacturing costs and potential challenges in mass production.

It is important to note that the advantages and disadvantages mentioned above are general considerations and may vary depending on the specific application and implementation of the foam-metal fin configuration in a PVT system.

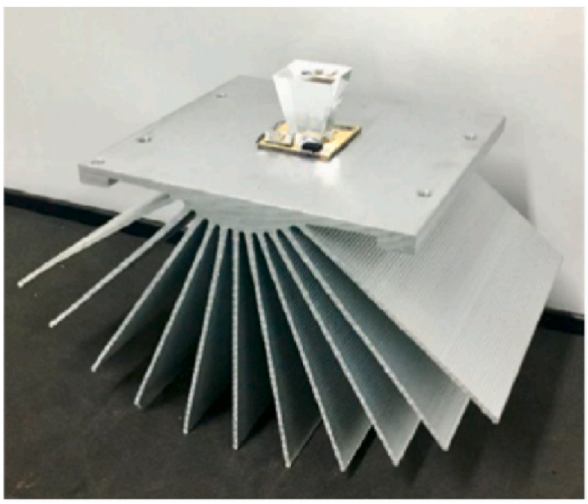


Fig. 17. Fins with open slots configuration [190].

The study in Ref. [190] examined natural convection heat transfer in metal foam heat sinks with open slots (Fig. 17). A heat sink's performance was tested by varying slot heights and orientations. Heat sink efficiency was significantly affected by the orientation of the slots in the experiment. The vertical orientation of the slot enhanced heat transfer, while the horizontal orientation reduced it. Heat transfer performance was also improved by increasing slot height. Using the results of this study, it may be possible to optimize the design of metal foam heat sinks for improved performance by understanding natural convection in them.

4.21. Flared-fin configuration

Flared-fin configuration refers to a specific design of heat transfer fins used in photovoltaic/thermal (PVT) systems. PVT systems combine the generation of electricity from solar panels with the extraction of heat from the panels to create a dual-purpose system [191].

Advantages of using a flared-fin configuration in PVT systems include [192,193]:

Enhanced heat transfers: The flared shape of the fins increases the surface area for heat transfer. The wider base of the fins allows for more effective dissipation of heat from the PV cells to the surrounding air or fluid. This results in improved thermal performance, lower operating temperatures, and potentially increased electrical efficiency of the PV cells.

Improved airflow distribution: The flared design helps in directing and distributing the airflow more evenly over the surface of the PV cells. This reduces the occurrence of hot spots and ensures better cooling uniformity, leading to improved overall system performance and longevity of the PV cells.

Reduced pressure drop: The gradually widening shape of the fins in a flared-fin configuration can help reduce pressure drop across the system. The smooth expansion of the fin geometry minimizes flow resistance, resulting in lower energy consumption requirements for fans or pumps used to maintain the desired airflow rate.

Disadvantages of using a flared-fin configuration in PVT systems include [194,195]:

Manufacturing complexity: The design and manufacturing of flared fins can be more complex compared to simpler fin configurations. The flared shape requires precise fabrication techniques to ensure consistent dimensions and proper alignment. This complexity can increase production costs and pose challenges in mass production.

Potential space constraints: The flared shape of the fins may require additional space compared to other fin configurations. This can be a limitation in installations where space is limited, such as rooftop installations or compact PVT systems. Proper consideration of available space and system design constraints is necessary when opting for a flared-fin configuration.

Maintenance challenges: The flared design can make cleaning and maintenance more challenging compared to simpler fin configurations. The wider base and tapered shape may create pockets or crevices where dust, debris, or other contaminants can accumulate. Regular cleaning and maintenance procedures must be followed to prevent reduced heat transfer efficiency due to fouling.

It is important to note that the advantages and disadvantages mentioned above are general considerations and may vary depending on the specific application and implementation of the flared-fin configuration in a PVT system. Proper system design and optimization are essential to maximize the benefits of this fin configuration.

An experimental investigation of flared-fin heat sinks' heat dissipation performance was presented by Luo [86], on "Concentrated Photovoltaic Modules" (CPV). Various flared-fin heat sinks were tested to compare their heat dissipation efficiency. By using flared-fin designs, the heat sinks were found to be significantly more efficient at dissipating heat. As the number of fins and the width of the fins increased, the heat dissipation performance of the heat sinks improved significantly. Using flared-fin heat sinks for CPV modules significantly improves their heat dissipation performance and the performance increases with more fins and wider fins, according to this paper.

5. Mathematical models of fins design

Various mathematical models are used to design fins, which are based on a variety of performance parameters. For example, the following models have been proposed: Fin efficiency models [196], fin heat transfer models [197], fin temperature rise models [198], fin thermal conductivity models [199], fin area models [200], fin geometry models [201], fin flow models [201], fin air velocity models [202], fin pressure drop models [203], and fin surface temperature models [204].

The equations used to evaluate the heat rate and fin efficiency of fins are [205]:

$$\text{Heat Rate: } HR = (\Delta T \times A_f \times h) \,/\, (P \times \eta_e) \tag{6}$$

$$\text{Fin Efficiency: } \eta_f = (h \times A_f \times \Delta T) \,/\, (P \times \eta_e \times (1 + h \times L)) \tag{7}$$

$$\text{Fourier's Law of Heat Transfer: } Q = (k \times A_f \times \Delta T) \,/\, L \tag{8}$$

where: "ΔT = temperature difference, A_f = fin area, h = heat transfer coefficient, k = thermal conductivity, Q = heat transfer rate, and L = fin length".

6. Comparison and critical review

An examination of the parameters affecting the selection of PV cooling fins is presented in this section.

- **Fin weight:** Choosing a PV system elevation requires consideration of fin weight. Adding additional support is likely to be necessary when installing heavier fins. Heavy fins are also more expensive to install due to the need for more materials and labour [206].

- **Fin cost:** In choosing a fin or PV system elevation, fin cost is an important factor to consider. Fin costs are determined by the fin's size, material, and complexity. The price of heavier fins is higher than that of lighter fins, as well as the cost of more complex fins is higher than that of simpler fins [206].
- **Fin and PV system elevation:** Choosing a fin or PV system elevation requires consideration of the fin or PV system height. It is usually more expensive and more difficult to construct higher elevations [207].
- **Fin and wind speed:** In choosing an elevation for a fin or PV system, fin and wind speed must be considered. There may be an increase in wear and tear on the system due to higher fin speeds and higher wind speeds, which should be taken into account when designing the system [208].
- **Fin Geometry:** The fin geometry of a heat sink determines how it is shaped. In addition to rectangular and circular fins, elliptical fins can also be used. There are advantages and disadvantages to each fin geometry. It is most common to use rectangular fins. Surface area, manufacturing ease, and pressure drop all make them the best. Because of the limited fin height, it is not as effective as it could be in transferring heat. It is easier to manufacture circular fins, but they have a higher pressure drop and are more difficult to manufacture. Circular fins provide the best heat transfer performance. The fin spacing of these fins is also greater than that of rectangular fins. Circular and rectangular fins are balanced in terms of heat transfer performance and pressure drop by elliptical fins. The process of manufacturing them is also easier than that of circular fins [209]. In their paper, Sadeghianjahromi and Wang [209] reviewed empirical and numerical studies on fin-tube heat exchangers and their heat transfer optimization mechanisms in detail. Compared to this, the present study examined all types of heat exchangers that use fins for heat transfer in detail.
- **Fin Uniform Temperature Distribution:** Heat transfer is more efficient when fin temperatures are uniform. Heat can be transferred from the entire surface area of the fins by exposing them to the same temperature. When heat sources are not distributed evenly, this is particularly important [210].
- **Symmetrical fin arrangements:** Fins are uniformly sized and shaped along their length. As a result, cooling efficiency is consistently high. When the exact cooling requirements are unknown or when cost is the primary concern, symmetrical fins are often used. There are various sizes and shapes of fins along the length of an asymmetrical fin arrangement. A more flexible cooling system can be tailored to meet specific requirements and thus provide more flexibility in terms of cooling performance. Fins with asymmetrical shapes can provide a better cooling performance than those with symmetrical shapes, but they can be more difficult to manufacture. Asymmetrical fins are generally thinner than symmetrical fins in terms of fin thickness. As a result, symmetrical fins dissipate heat more effectively and keep components cooler. The downside is that thicker fins are costlier to manufacture and harder to install. When it comes to fin arrangements and fin thickness, there is no one-size-fits-all solution. Depending on the type of application and the cooling requirements, the choice depends on which type to use. Fins with symmetrical shapes are more economical, while fins with asymmetrical shapes offer better cooling efficiency [211].
- **Fin thickness: Watercraft or vessels should consider fin thickness when designing them.** Performance can vary greatly depending on the fin thickness. The thinner the fins, the faster the vessel will accelerate and the higher the top speed will be. As thin fins can be more easily bent or broken when encountering debris or shorelines, they can be more susceptible to damage. In choppy waters, thicker fins contribute to smoother handling because they provide greater stability. Their greater resistance to debris and shoreline damage also makes them a better protection against debris and shorelines. As a result of thicker fins, the vessel's performance can be negatively affected, resulting in slower acceleration and top speeds. Therefore, the purpose of the vessel will ultimately determine the fin thickness. Fins with thinner walls are best if performance is a priority. However, thicker fins are a better choice if durability and stability are more important [212].

7. Conclusion

Based on the type of module and arrangement of the system, this review examined the various cooling techniques that are suitable for PV modules. A wide range of active and passive cooling approaches including air, water, fins, heat sinks, and phase change materials are classified, characterized, and evaluated. There is a greater focus on fin types, design, and modelling in this paper. Further, the suitability of the methods and the results of studies that utilized them were considered, as well as ways to enhance the effectiveness and durability of the components. According to the study, fin configurations have a significant effect on both the negative and positive thermal performance of the fins, especially on the pressure drop, their designs, weight, quantity, and type of metal used. Different fin configurations improved PV performance and efficiency. Depending on the fin configuration, an improvement rate of 8%–10% can be obtained.

For the next generation of PV technologies and future studies, a comparison of fin-based PV cooling techniques was conducted. The suggestions made will benefit upcoming research and engineers/specialists working on PV design, engineering, and application. A road map for future research and development can be derived from each of the critical findings in the article. The following points outline potential directions for future studies in the field:

- Experimental validation: while the current review extensively analyses and evaluates various cooling techniques, there is a need for more experimental validation of the proposed methods. Conducting real-world experiments with different fin configurations, materials, and designs would provide valuable data to verify the theoretical findings and enhance the credibility of the research.
- Long-term performance and durability: future studies should focus on assessing the long-term performance and durability of the cooling systems. Evaluating the reliability and stability of fin-based cooling techniques over extended periods will provide insights into their effectiveness and potential issues that may arise over time.

- Optimization of fin design: Further research can be conducted to optimize the design of fins for enhanced PV performance. Exploring innovative fin geometries, such as novel shapes or patterns, and conducting computational simulations or optimization algorithms can help identify the most efficient fin configurations in terms of heat transfer and pressure drop.
- Cost analysis and economic viability: it would be valuable to conduct a comprehensive cost analysis of the different cooling techniques to assess their economic viability and practical implementation. Considering factors such as manufacturing costs, maintenance requirements, and energy savings achieved through improved PV performance, a detailed economic analysis would assist in determining the most cost-effective cooling approach for different applications.
- Hybrid cooling approaches: exploring hybrid cooling approaches that combine multiple techniques, such as air and liquid cooling or phase change materials with fins, could potentially offer synergistic benefits. Investigating the combined effects of different cooling mechanisms and their interactions could lead to more efficient and effective cooling strategies for PV modules.

Author statement

We would like to submit the following author statement in conjunction with our revised manuscript titled "Enhancement of Photovoltaic Module Performance using Fins for Cooling: A Comprehensive Review" for publication in Case Studies in Thermal Engineering.

First and foremost, we would like to express our gratitude to the editorial office and the reviewers for their time, expertise, and constructive feedback during the review process. Their insightful comments and suggestions have greatly contributed to the improvement and quality of our manuscript. The research presented in this manuscript aims to provide a comprehensive review of the use of fins for enhancing the performance of photovoltaic (PV) modules through cooling. We have thoroughly addressed the comments and suggestions made by the reviewers, ensuring that all their concerns have been carefully considered and incorporated into the revised manuscript. Throughout the revision process, we have made significant changes to the manuscript based on the feedback received. We have reorganized the structure of the paper to provide a clear and logical flow of information, addressing the temperature dependency of PV efficiency, introducing different cooling approaches and fin types, and presenting the results of our literature review on applied fins.

Furthermore, we have provided an overview of existing reviews on PV cooling in the introduction, allowing readers to better understand the state of the art and the distinctiveness of our review. We have also included a more detailed description and comparison of the cited papers in Table 1, addressing the need for further clarity in our previous submission.

In response to the reviewers' comments, we have ensured that all figures in the manuscript now include proper sources and citations. We have improved the conclusions section to provide more quantitative findings and have added a future research perspective to guide further investigations in this field. Additionally, we have thoroughly proofread the manuscript to correct minor English mistakes and ensure the clarity of the language used.

We would like to reaffirm our commitment to the ethical standards of academic publishing. The manuscript submitted is original work, and we have appropriately cited and referenced all sources used. We confirm that there are no conflicts of interest related to this research.

Once again, we express our sincere appreciation for the valuable feedback provided by the reviewers and the editorial team. We believe that the revisions made have significantly strengthened the manuscript and its contribution to the field of thermal engineering. We hope that our revised manuscript meets the requirements of Case Studies in Thermal Engineering and kindly request its reconsideration for publication.

Declaration of competing interest

The authors declare no conflict of interest regarding the publication of this article. The content and findings presented in this paper are based on objective research, scientific evidence, and the authors' expertise in the field. There are no financial, professional, or personal relationships that could potentially bias the content or interpretation of the information presented. The authors have no affiliations with organizations or entities that may have a vested interest in the subject matter discussed. This declaration ensures the integrity and impartiality of the information provided in this review article.

Data availability

No data was used for the research described in the article.

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